

What is the OSI model?

The Open Systems Interconnection (OSI) Model is a description of how the Internet works. It breaks down the functions involved in sending data over the Internet into seven layers. Each layer has some function that prepares the data to be sent over wires, cables, and radio waves as a series of bits.

The seven layers of the OSI model are:

1. Application layer: Data generated by and usable by software applications. The main protocol used at this layer is [HTTP](#).
2. Presentation layer: Data is translated into a form the application can accept. Some authorities consider [HTTPS](#) encryption and decryption to take place at this layer.
3. Session layer: Controls connections between computers (this can also be handled at layer 4 by the [TCP protocol](#)).
4. Transport layer: Provides the means for transmitting data between the two connected parties, as well as controlling the quality of service. The main protocols used here are TCP and [UDP](#).
5. Network layer: Handles the routing and sending of data between different networks. The most important protocols at this layer are IP, ICMP, ARP, and BGP.
6. Data link layer: Handles communications between devices on the same network. If layer 3 is like the address on a piece of mail, then layer 2 is like indicating the office number or apartment number at that address. Ethernet is the protocol most used here.
7. Physical layer: Packets are converted into electrical, radio, or optical pulses and transmitted as bits (the smallest possible units of information) over wires, radio waves, or cables.